

Sepsis Digital Twin - Full Project Overview

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1) Executive Summary

This system predicts sepsis risk 4-6 hours before onset using ICU time-series data. It combines a strong tabular baseline (Histogram-based Gradient Boosting, HGB) with four deep learning models (GRU-D, LSTM, CNN-LSTM, Transformer) trained on real ICU data, and exposes results via a Streamlit dashboard and a FastAPI microservice. The dashboard shows per-model risk, an ensemble score, an agreement indicator, and detailed evaluation plots. The evaluator produces ROC/PR figures and a model comparison table with operating-point metrics at Recall ~0.85.

2) Quick Start (Local Dev)

- Dashboard: ``streamlit run dashboard_real.py``

- API:

3) Architecture Overview

- Frontend: Streamlit app (``dashboard_real.py``) with tabs for Risk Overview, Model Predictions, Risk Trajectory, Explainability, Performance, Clinical Workflow, Fairness, and exports.

- Back

4) Data Flow (End-to-End)

1. Acquire latest patient record from ``Dataset.csv`` or manual form entry.

2. Inte

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5) Models and Rationale

- Baseline (HGB): strong tabular performance, fast, interpretable, easy to calibrate; used as a sanity floor and fallback.

- GRU

6) Integration Layer (`integration_real.py`)

- Single adapter for baseline and DL inference keeps evaluator and dashboard consistent.

- GRU

7) Evaluation (`Capstone/evaluate_models.py`)

- Computes AUROC, AUPRC, default-threshold metrics, and operating-point metrics at Recall ~0.85:

- Accu

8) Dashboard (`dashboard_real.py`)

- Risk Overview: ensemble score, risk level, agreement metric, recommended action.

- Mod

9) Live Models in the Dashboard

- Uses up to 5 models live if artifacts exist: GRU-D, LSTM, CNN-LSTM, Transformer (all trained on real data), and HGB.

- Eval

10) Roles and Deliverables

- Person A (Data Engineering): ICU dataset (``Dataset.csv``), patient-wise splits, missingness strategy, ``outputs/config.json`` with feature/sequence settings.

- Pe
`mode

11) Recent Enhancements (This Review)

- Added operating-point metrics at Recall ~0.85 and exposed them in the dashboard.

- Align

12) Failure and Fallback Behavior

- Missing DL checkpoint: model skipped or runs with default weights; clearly logged.

- Miss

13) Deployment Notes

- Docker targets for API-only, dashboard-only, or complete system (API + dashboard).

- Heal

14) Demo Script (2 Minutes)

1. Open dashboard (<http://localhost:8501>). Show Risk Overview (ensemble, agreement, recommended action).

2. Sho

15) FAQ (Short)

- Why HGB baseline? Strong tabular baseline; sanity floor; fast and interpretable; fallback if DL missing.

- Why

Appendix: Model-Specific Details

GRU-D (handles missingness and time explicitly)

- Inpu

LSTM (recurrent memory)

- Inpu
imple

CNN-LSTM (local patterns + sequence memory)

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- Input

Transformer (self-attention)

- Input

Calibration and thresholds (how probabilities become alerts)

- Calibration
quality

Feature scope actually used

- Vital